

AKHIL MM *FULL STACK DEVELOPER*

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ABOUT ME

Full-stack developer with 1.5+ years of hands-on experience building real-world web applications and backend systems. Developed projects including a ride-booking platform, secure chat application with AI features, e-commerce platform, and accessibility-focused solutions. Interested in scalable system design, real-time applications, and solving practical problems through software. Strong focus on writing maintainable code, learning continuously, and building products with real-world impact.

Technologies

Frontend: HTML, CSS, JavaScript (ES6+), React.js, Redux, Tailwind CSS

Backend: Node.js, Express.js, REST APIs, Typescript

Database: MongoDB, NoSQL

Version Control: Git, GitHub

Cloud & Deployment: AWS EC2, AWS S3, Google Cloud Platform (GCP), Nginx

Tools & Version Control: Git, GitHub, Postman

Projects

Sarathi – Ride-Booking Platform

Sarathi is a specialized driver-on-demand platform that allows vehicle owners to find and hire nearby professional drivers to operate their personal cars. The application enables users to locate available drivers in their vicinity, book specific time slots for their trips, and process payments securely within the app once the journey is complete. It features an intuitive interface for owners to manage their vehicles and bookings, a dedicated driver portal for trip fulfillment, and a robust admin panel for verifying driver credentials like licenses and Aadhaar. By bridging the gap between car owners and verified drivers, Sarathi provides a convenient and safe solution for personalized transportation services.

- **Technologies Used:** TypeScript, Node.js, Express, MongoDB, React, material ui, TSyringe (Dependency Injection), React, Tailwind CSS
- Built the backend in TypeScript with Express, MongoDB, and TSyringe for dependency injection following SOLID principles.
- Integrated OTP email verification via Nodemailer for both user and driver registration. • Used Upstash Redis to temporarily store registration data and Socket.IO  connection details for real-time features.
- After registration, drivers submit vehicle image and RC book image, followed by license, Aadhaar, and other personal details.
- All images are securely uploaded to Cloudinary using signed upload with server-generated signatures.
- Implemented Stripe connected account onboarding for drivers and automatic fare split (90% driver / 10% platform).
- Used Google Maps API for location autocomplete and roadwise distance calculation between pickup and destination
- Developed a full chat and video calling feature using raw WebRTC and Socket.IO  for real-time communication. Chat supports text, emojis, images, PDF documents, and file sharing.
- Frontend built with React, TypeScript, and Material UI (MUI) with responsive design.
- Hosted backend on AWS EC2 using PM2 + NGINX, and deployed frontend via Vercel

AI-Powered Real-Time GroupChat Platform

Summary: Built a scalable real-time group chat application with an integrated Gemini AI Assistant using a Retrieval-Augmented Generation (RAG) architecture. Designed with Clean Architecture principles to support secure media sharing, semantic AI context retrieval, and high-performance rendering for large chat histories.

Technical Stack & Key Features:

- **AI & RAG Architecture:** Integrated Google Gemini API with a custom RAG pipeline using MongoDB Atlas Vector Search and gemini-embedding-2. Combined semantic retrieval with chronological chat history to generate context-aware AI responses.
- **High-Performance Frontend:** Implemented virtualized rendering using TanStack Virtual for smooth infinite scrolling across thousands of messages. Optimized rendering with dynamic height measurement, React.memo, and stable callback patterns to minimize unnecessary re-renders and memory usage.
- **Backend Architecture:** Built the backend using TypeScript and Clean Architecture principles (Use Cases, Repositories, Services, Controllers) for scalability, maintainability, and separation of concerns.
- **Security, Cloud & Deployment:** Deployed the backend on a **Google Cloud Platform (GCP)** instance and the frontend on **Vercel** for high availability. Implemented JWT access/refresh token rotation and Google OAuth 2.0. Integrated AWS S3 presigned URLs for direct media uploads, drastically reducing server bandwidth usage.
- **Real-Time Features & UX:** Developed a responsive UI using React, Tailwind CSS, and Framer Motion with Socket.IO  -powered realtime messaging, IntersectionObserver-based read receipts, intelligent message grouping, and AI-assisted chat interactions.

EYEGAZE | Full-Stack E-commerce Platform

Technologies Used: Node.js, Express.js, MongoDB, EJS, Bootstrap, Chart.js, HTML5, CSS3

- **Engineered** a comprehensive eyewear e-commerce platform featuring dynamic product discovery, persistent cart management, and a seamless multi-step checkout flow.
- **Architected** a premium, high-performance Admin Dashboard using **Chart.js** to visualize real-time business metrics like Total Revenue, Average Order Value (AOV), and User Growth trends.
- **Integrated** secure payment gateways via **Razorpay API** and streamlined product catalog management using **Multer** for optimized multi-image processing.
- **Implemented** robust session-based authentication and specialized middleware to ensure secure user authorization and data protection across the platform.
- **Developed** advanced management modules for products, categories, and inventory, alongside a dynamic **Offer and Coupon system** designed to maximize sales conversions.

Education

Bachelor of Science in Mathematics | 2019 – 2022 University of Calicut